

Viyom Shukla

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GitHub: github.com/viyomshukla

TUF+: 350+ DSA problems | LeetCode: 200+ problems

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Portfolio: viyom-potfolio.vercel.app

Location: Noida, India

EDUCATION

- **Jaypee Institute of Information Technology** Noida, India
Integrated B.Tech + M.Tech in Computer Science Engineering; CGPA: 7.62/10 2022 – 2027
- **Gayatri Public School** Agra, India
Class XII (CBSE): 88.6% | Class X (CBSE): 86.6%

EXPERIENCE

- **Distance Connect** Remote
Full Stack Intern — Backend (Python) & Frontend Aug 2025 – Oct 2025
 - **End-to-End Feature Ownership:** Designed and developed platform features end-to-end for an early-stage EdTech startup connecting students with IIT and AIIMS mentors, building Python backend services and the REST APIs consumed by the client layer.
 - **Python Backend Service:** Engineered and deployed a reliable, scalable Python backend service powering an AI conversational interface, automating student onboarding and query resolution to reduce manual support overhead.
 - **Third-Party REST API Integration:** Integrated the Google Meet REST API to build a secure 1:1 video consultation flow end-to-end, enabling virtual mentoring sessions directly within the platform.
- **Student Placement Committee, IIIT Noida** Noida, India
Coordinator Aug 2024 – Jan 2025
 - **Corporate Outreach:** Coordinated with the placement head and cross-functional student teams to source companies actively hiring for campus roles, compiling job requirements and eligibility criteria to prioritize outreach efforts.

PROJECTS

- **Thumblify — AI-Powered Thumbnail Generation & Editing Platform (React.js, Node.js, MongoDB, REST APIs, Cloudinary)** *Live Demo*
 - **Scalable, Fault-Tolerant Backend:** Designed a reliable backend pipeline routing requests across multiple LLM providers via Replicate REST APIs with provider-level fallback, sustaining 100+ users generating 2K/4K thumbnails without service interruption.
 - **Payments & Transaction Integrity:** Eliminated transaction-tampering risk in a credit-based subscription and billing system by implementing server-side Razorpay webhook signature verification, achieving 100% payment integrity across all completed transactions.
 - **Low-Latency Multi-Channel Access:** Reduced generation friction for mobile users by building a Twilio-integrated WhatsApp bot alongside an in-app chatbot, enabling prompt-to-thumbnail generation without requiring users to open the web dashboard.
- **NextTrip — Booking & Payments Platform (Backend: Node.js, MongoDB, REST APIs)** *Source Code*
 - **Secure REST API Design:** Engineered role-based access control with Google OAuth 2.0 and JWT across admin/traveler roles, securing every REST API route, and automated Razorpay payment capture with server-side verification, reducing manual reconciliation to 0.
 - **KYC & Identity Verification:** Built an Aadhaar document-upload and admin-review flow that gates organizer onboarding, applying KYC-style verification before a user can publish trips and take bookings.
 - **Data Modeling & Query Performance:** Designed a normalized 6-collection MongoDB schema with indexed queries to serve trip listings and booking analytics, and built the admin dashboard on top of it.
- **DeepMood — GAN-Based Class-Imbalance Correction (Python, Machine Learning, Generative AI)** *Notebook*
 - **Data Preprocessing & EDA:** Analyzed a real-world depression dataset through exploratory data analysis, selected the most relevant features, and normalized the data to prepare it for model training.
 - **Synthetic Data Generation with CGAN:** Identified that depression cases were underrepresented in the dataset, then built a Conditional GAN to generate 5,000+ synthetic records, balancing the classes and improving classifier accuracy to 92%.

RESEARCH & PUBLICATIONS

- **Enhancing Network Intrusion Detection Using Conditional GANs for Imbalanced Data Classification** *IEEE Xplore*
 - Implemented a Conditional GAN to synthesize rare attack samples in the NSL-KDD dataset, balancing class distribution and training Random Forest, SVM, and Neural Network classifiers to achieve 99.8% detection accuracy across underrepresented threat categories.

ACHIEVEMENTS

- **International Conference Speaker:** Presented published research at the 17th IEEE IC3-2025, IIIT Noida. *Certificate*
- **Competitive Programming:** Solved 450+ DSA and 50+ SQL problems across TUF+, LeetCode, and GeeksforGeeks, focusing on optimization, time complexity, and algorithm design.
- **Smart India Hackathon (SIH '23):** Selected among the top teams across India in the Tourism domain.
- **National Hackathon Competitor:** Competed in the Amazon ML Challenge and Adobe India Hackathon '25.

TECHNICAL SKILLS

Programming Languages: C++ (Proficient), Python (Proficient), JavaScript, C, Golang(Basic) ,SQL

Backend & APIs: Node.js, Express.js, REST APIs, JWT, OAuth 2.0, Webhooks

Databases: MongoDB, MySQL

Tools & Platforms: Git, GitHub, AWS (basic), Postman, Firebase, Cloudinary, React.js, Tailwind CSS

Core CS: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Machine Learning